# Soumil Iyer

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## EDUCATION

**Purdue University, West Lafayette, IN**

Master of Science in Mechanical Engineering (GPA: 4.00) — January 2026 - December 2026 (Expected)
Bachelor of Science in Mechanical Engineering (GPA: 3.38) — August 2022 - December 2025

## EXPERIENCE

**Rapid Prototyping Lab Student Employee | Purdue University** — **August 2026 - Present**

- Guiding CAD-to-part FDM/FFF workflow in Bambu/Cura, covering printer/material selection, orientation, supports, infill, layers, and temperature
- Set up and calibrate printers; perform preventive maintenance and troubleshoot process/equipment issues to sustain reliability and consistency
- Preflight geometry/slicer/material failure risks; coach CAD/process changes improving DFM, strength, finish, repeatability, and waste
- Isolate failures across geometry, slicer settings, material behavior, and machine condition to reduce failed prints, rework, and waste

**Graduate Teaching Assistant - Intro to Manufacturing | Purdue University** — **January 2026 - May 2026**

- Instructed 19 students in lathing, milling, CNC, metal, wood, electronics, assembly, and DFM labs from CAD intent through finished parts
- Taught Y14.5 GD&T/tolerancing and metrology; reviewed drawings, BOMs, and assembly procedures against specs for manufacturability and fit
- Supervised machine use, inspections, safety, and design validation while troubleshooting fabrication, dimensional, and assembly issues

**M.E. Undergraduate Research Intern | Purdue University** — **June 2025 - August 2025**

- Manufactured 60+ dogbones and 24 disks across 6 thermoplastics in horizontal/vertical orientations for controlled material/process comparison
- Performed ASTM D638 tensile tests and ultrasonic NDE to quantify anisotropy and destructive versus nondestructive material response
- Processed stress-strain and wave-speed data in MATLAB to calculate Young's modulus and compare failure behavior across materials/orientations
- Documented fabrication, test methods, uncertainty, and results showing build orientation significantly affected tensile modulus and failure behavior

**M.E. Undergraduate Research Assistant - Energy Storage Analytics | Purdue University** — **August 2024 - December 2024**

- Built a Python/customtkinter tool with file upload, analysis, and plotting workflows to process and visualize 1,000+ Li-ion battery datapoints
- Created reusable comparison/visualization workflows for repeatable data trending, performance interpretation, and technical documentation
- Interpreted Li-ion performance trends against cell design, behavior, and optimization to strengthen battery-focused engineering analysis

**Teaching Assistant - Thermodynamics I & ENGR 131 | Purdue University** — **August 2024 - December 2024**

## PROJECTS

**Design of Experiments - Lead-Free Solder Joint Process Optimization | IE 533** — **March 2026 - April 2026**

- Designed randomized 2^3 Minitab DOE for SAC305 shear strength using 230/260 C, 0.5/1.0 mg paste, and 1/4 C/s cooling; 16 simulated runs
- Used 2 replicates; controlled PCB/pad geometry, 120 C preheat, ambient conditions, and equipment; randomized order and achieved ~0.83 power
- Applied GLM/ANOVA (p=0.006), Pareto/main-effect/interaction plots, and residual diagnostics; all main effects significant, interactions NS
- Quantified effects: temp +6.90, cooling +5.46, paste +3.36 N/mm^2; recommended 260 C, 1.0 mg, 4 C/s while documenting simulated-data limits

**Mechanical Engineering Senior Design - Assisted Eating Utensil** — **August 2025 - December 2025**

- Led CAD/FDM prototyping for a sub-$25 self-leveling utensil; translated user needs into requirements and iterated the bearing mechanism
- Turned user feedback into ribbed grip, retention strap, and detachable steel-head changes while balancing usability, assembly, and DFM
- Iterated 3D-printed prototypes through fit, handling, manufacturability, and material/geometry tradeoffs before final design freeze
- Validated with 2 m drops, dishwasher cycles, and insertion/removal-force tests while maintaining function, durability, and usability

**Autonomous Robot Control & Line Following | ME 375** — **April 2025 - May 2025**

- Characterized motors/encoders and sensors: K=14.375/14.95, tau=0.10/0.08 s, 5.13 in/s at 100% PWM, IR error <0.5 cm over 10-80 cm
- Built Simulink state logic for rest/race/recovery/parking with PID/PI, encoder feedback, line/IR sensing, and static-friction compensation
- Developed a Simulink root-locus wheel-speed controller and validated integrated hardware through two laps and a 30 cm parking maneuver

**Heat Transfer Analysis - Blanket Insulation Materials | ME 315** — **March 2025 - April 2025**

- Instrumented polyester/wool/down with thermocouples/LabVIEW; logged 1-s data for 1-hour cooling runs from 37 C toward ~5 C
- Verified lumped-capacitance validity at Bi ~0.03; linearized transients and calculated k=0.390/0.0152/0.0344 W/m-K for polyester/wool/down

**Wind Tunnel Aerodynamic Analysis - Race Car Spoiler | ME 30801** — **March 2025 - April 2025**

- Designed/FDM-printed a SolidWorks spoiler; calibrated LVDT/load-cell setup and tested 0-60 deg AoA at 30.880 m/s with 3 trials/angle
- Processed 15-s datasets in MATLAB using averages/error bars and Cd/Cl; identified peak Cl ~0.65 near 40 deg and documented ~2 mm print shift

**Autonomous Rover Development - Drone Video VIP | VIP 27920** — **August 2023 - December 2023**

- Worked on Purdue Drone Video VIP rover/hardware team to improve autonomous navigation and hardware/software integration
- Supported autonomous rover sensing/navigation using visual/object detection, IR, LiDAR, and hardware/software integration

## SKILLS & CERTIFICATIONS

**CAD & Design:** CATIA V5/V6, Siemens NX (PLM/PDM), SolidWorks, Fusion 360, AutoCAD, Mastercam, tolerance stack-up
**Mechanical Analysis:** Solid mechanics, mechanics of materials, structural/thermal FEA, heat transfer (ME 323/32301/315/354)
**Design & Reliability:** GD&T (ANSI/ASME Y14.5), DFM/DFA, engineering drawings, BOMs, DFMEA, Fault Tree Analysis (FTA)
**Materials:** Engineering metals/polymers, structure-property relationships, material selection, ASTM D638 characterization (MSE 230; research)
**Manufacturing Processes:** CNC/manual machining, FDM/FFF; stamping, deep drawing, injection molding, heat treating, welding fundamentals
**Testing & Data:** DoE, ANOVA/GLM, Minitab, MATLAB, LabVIEW, DAQ, metrology, thermocouples, LVDTs, NDE, uncertainty analysis
**Programming & Controls:** Python, MATLAB, C, Simulink, ROBOGUIDE, Studio 5000, Ladder Logic, PID/PI, root locus, sensor calibration
**Lean & Manufacturing Systems:** Lean, Kaizen, 6S, Pareto analysis, 5 Why, Fishbone diagrams
**Certifications:** SolidWorks Milestones (in progress); SOLIDWORKS certification (exam pending); FEA; Simulink; MATLAB; Python; Arduino